VoltTrace Digital Battery Passport

COMPREHENSIVE AUDIT REPORT • OVERVIEW & HEALTH SUMMARY

Battery Pack: VT-1001 • Standard: Policy VT-202 (EU 2023/1542)

VT-1001

STATUS: ACTIVE

86%

HEALTH: GOOD

MEASURED CAPACITY

64.5 kWh / 75 kWh

VOLTAGE / CYCLE COUNT

400 V • 620 cycles

CIRCULARITY DISPOSITION

SECOND_LIFE_READY

& Attention Required: 1 unresolved issue(s) recorded on this battery pack.

1. Battery Specifications & Identity Matrix

Manufacturer	**Northvolt AB**	Model / Part Number	**VoltCore NMC 75**
Cell Chemistry	**NMC**	Nominal Voltage	**400 V**
Rated Pack Capacity	**75 kWh**	Manufacturing Date	**10 Apr 2022**
Current Fleet / Owner	**Zypp Electric Fleet**	Registration Date	**12 Apr 2022**
Vehicle Application	**Electric 2W / Fleet**	Passport Ledger State	**ACTIVE**

2. Registry Activity Snapshot

LIFECYCLE EVENTS
2
Recorded on ledger

DIAGNOSTIC TESTS
1
Impedance benchmarks

CERTIFICATE DOCS
1
Bit-for-bit verified

WORK ORDERS
1
0 open orders

3. Mandatory Service Work Orders (Policy VT-202 Enforcement)

Real-world service software prevents closure without verified technician IDs, diagnostic report proof, and SHA-256 digital locking.

ORDER ID	OPEN DATE	SERVICE CENTER	TECHNICIAN	STATUS	HEALTH RECORDED
WO-2024-0512	12 May 2024	Zypp Mobility Central Hub …	TECH-409	**COMPLETED**	**86% SoH**
Action Taken: Attached TÜV Rheinland ISO-17025 report. Cell balancing within OEM specs.					

4. Chronological Lifecycle Passport (Event Audit Log)

TIMESTAMP	EVENT TITLE	SUBMITTING ORG	ACTOR / ROLE	VERIFICATION
12 May 2024	**Annual Diagnostic Inspection**	Zypp Hub	TECH-409	' VERIFIED
Impedance test completed within OEM spec.				
10 May 2023	**Fleet Registration**	Northvolt OEM	OEM Admin	' VERIFIED
Commissioned onto Zypp Bangalore logistics route.				

5. Laboratory Diagnostic Test Records & Impedance Telemetry

TEST DATE	TESTING ORGANIZATION	METHOD / SPEC	CAPACITY	RESISTANCE	RECORDED SOH
12 May 2024	TÜV Rheinland India	ISO-17025 Impedance Spec	64.5 kWh	1 4 . 2 m ©	86%

6. Attached Accreditation Documents & Merkle Checksums

FILE NAME	ISSUER	TRUST TIER	INTEGRITY STATUS	SHA-256 CHECKSUM
VT1001_TUV_Diagnostic_Report_2024…	TÜV Rheinland India	Tier 1 Accredited	' VERIFIED	a9f1b2c3d4…

7. Flagged Contradictions & Items Requiring Attention

[DISC_VALID_SIG_IMPOSSIBLE_SOH] Valid Signature, Physically Contradictory Degradation STATUS: UNRESOLVED • SEVERITY: CRITICAL

Report signed by accredited lab contains SoH 94% after 1,200 cycles, violating thermodynamic limit.

8. Physics Consistency Engine & Sub-Component Lineage

' Thermodynamic Fade Law	Validates degradation trajectory against C-rate, chemistry, and calendar aging limits.
' Odometer & Cycle Rollback Check	Ensures cycle counts never decrease and mileage correlates with BMS amp-hour throughput.
' Provenance Signature Proof	Blocks work order closure until accredited TÜV / lab diagnostic report hash is anchored.

DECENTRALIZED BATTERY PASSPORT AUTHENTICATION & CERTIFICATION SEAL

This document is generated by VoltTrace as an authoritative digital product passport.
Every historical diagnostic record, work order sign-off, and attached certificate is cryptographically locked.
Compliant with EU Battery Regulation 2023/1542 (Articles 65 & 77) and VoltTrace Mandatory Provenance Policy VT-202.

`Report Checksum SHA-256: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 • Verified at: 2026-10-30T21:20:34.681Z`